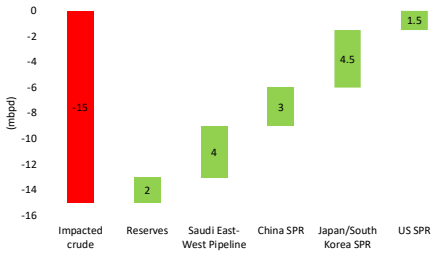
EXPERT - GLOBAL OIL, LNG, REFINING

SECTOR UPDATE

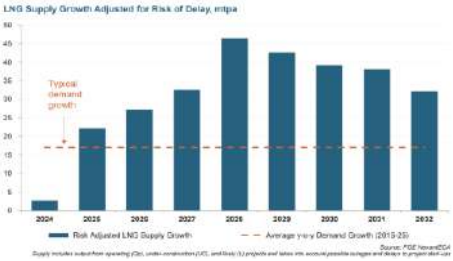
nuvama institutional equities

Oil to hit USD110–150 if long outage

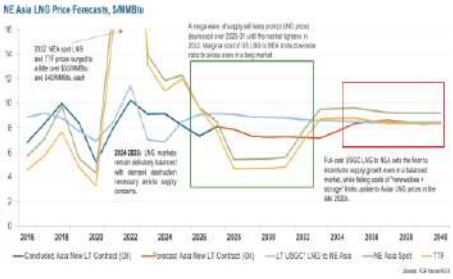
Strategic reserves bridge supply gap



LNG supply 2x demand for decade



LNG price to halve to USD6 by 2028



Strong GRM likely on low capacity adds

USD150/bbl oil in sight

We hosted FGE Nexant experts (Page 18) on global oil, refining, LNG outlook and India's high exposure to Strait of Hormuz (SoH) closure.

Takeaways: i) SoH's continued closure (~20mbpd) **risks crude at USD110–150/bbl in 4–8 weeks**; releasing strategic reserves near-term relief, may lead to future demand restocking. ii) Qatar LNG production to restart in two weeks of SoH reopening as facilities remain undamaged. 2028 onwards, Asian LNG prices **may fall to USD6/mmbtu given huge new supply.** iii) Refining strong on run cuts, tight supply. Crude flexibility key; **limited capacity adds to keep long-term margins robust.** iv) ***India highly vulnerable:*** Short term crude manageable; high LNG prices may cut imports; LPG most vulnerable.

High oil prices on continued outage; reserves offer near-term relief

Middle East (ME) escalation has disrupted ~20mbpd (~20% of global supply) through SoH. FGE estimates four/eight-week outage could push crude to USD110/150 per bbl, likely triggering demand destruction. Asia's (~13mbpd to China, India, Japan, South Korea) most affected, while strategic reserves cushion supply shocks; countries with limited reserves may cut refinery runs. Releasing 300–400mn bbl could provide near-term relief, but create future restocking demand. ME export limits could force 6–7mbpd (>200mn bbl) shut-ins in March, keeping crude market tight.

Asia supplies impacted; FGE forecasts LNG prices at USD6 by CY28E

FGE noted a potential SoH blockade could disrupt ~80mtpa of LNG from Qatar and Abu Dhabi, mainly affecting Asia (India, China, Taiwan, Pakistan, Bangladesh), with a daily loss of three cargoes (~230,000 ton). Despite attacks, ME LNG facilities remain operational, supporting supply recovery if undamaged. While spot prices rose, FGE expects them to ease as North America and ME supply emerges beyond 2027, with long-term prices converging at USD 8–10/mmbtu. Asia spot price could fall to USD6/mmbtu by CY28E (from USD16 now).

Strong refining margins on run cuts, tight supply; long term bright

FGE forecasts 2.5–3mbpd refinery run cuts in Mar-26, lowering Asian product flows by 10–15%, causing 600kbpd gasoline and 900kbpd gasoil losses into March/April. Asian margins rose USD5–7/bbl versus Feb-26; OECD (1bn bbl) and Europe's diesel stocks (180mn bbl) aid flow balance, though ATF premiums remain high. Few post-2026 capacity additions, rising demand support strong long term refining prospects.

India navigating through supply disruption; LPG most vulnerable

Crude: Bridgeable over the short-term through reserves and alternate sourcing. ***Refining:*** Margins strong on supply disruption; crude supply remains key. ***LNG***: High LNG prices shall import cuts, demand destruction. However, FGE sees India's long-term LNG demand growth intact. ***LPG:*** Most vulnerable as shortage may lead to cooking gas unavailability, likely restaurant shutdown and demand destruction, which could last six–eight weeks.

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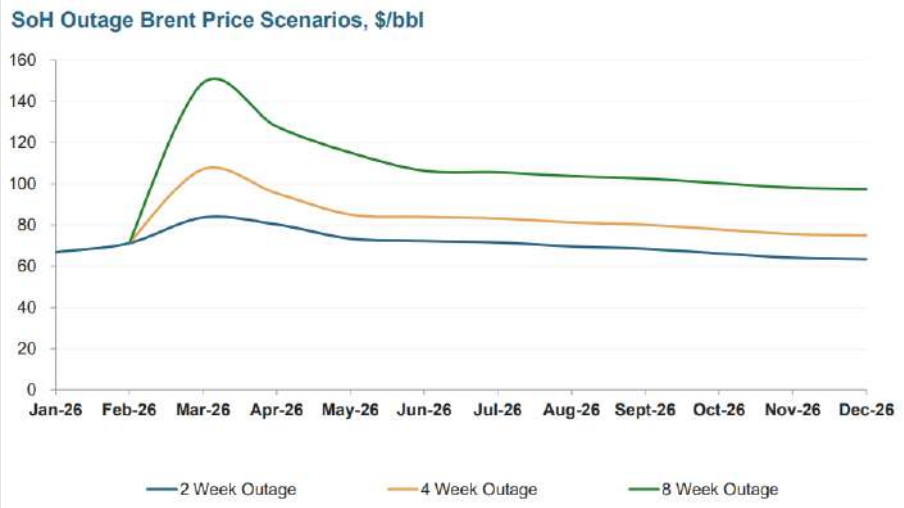
Oil: ME conflict disrupts crude flows

Escalation of the Middle East conflict has significantly disrupted crude flows through the Strait of Hormuz, a critical chokepoint that carries ~20mbpd (~20% of global supply). Traffic has slowed to a near halt, tightening the global crude balance. The market initially priced in a two-week disruption but expectations are shifting toward longer outage. FGE estimates that a four-week disruption could push oil prices to ~USD110/bbl, while an eight-week outage could see prices approach ~USD150/bbl, though such levels would likely trigger demand destruction and alternative supply responses.

Exhibit 1: Iran war: Near-term risk remains skewed to the upside

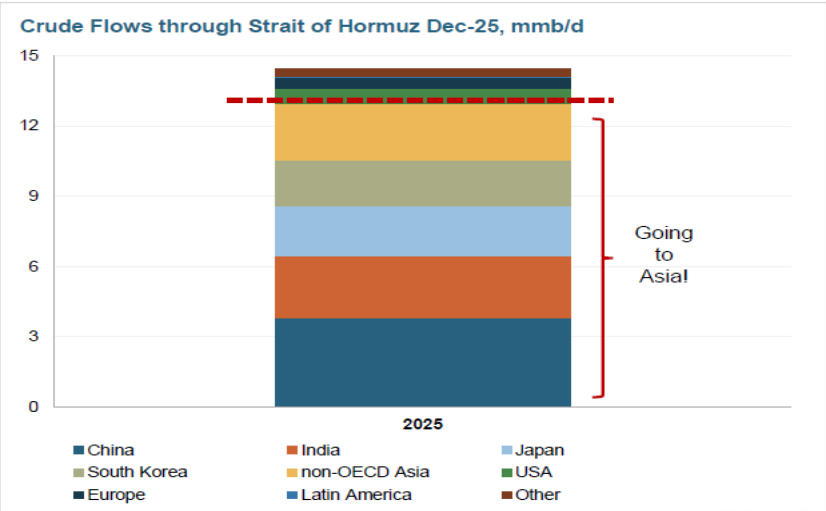


Source: FGE NexantECA

The Strait of Hormuz chaos

The disruption is largely Asia-centric, with ~13mbpd of Hormuz crude typically flowing to Asia—primarily China, India, Japan and South Korea—while Europe and the US remain relatively less exposed. Although many Asian buyers hold strategic petroleum reserves that can help cushion the immediate shock, countries with limited reserves are already announcing refinery run cuts.

Exhibit 2: Crude flow disruption is largely Asia centric



Source: FGE NexantECA

~15mbpd of crude was exported through Hormuz. These volumes are now stranded in the Persian Gulf, resulting in a bigger pull from Asia on Atlantic Basin crude.

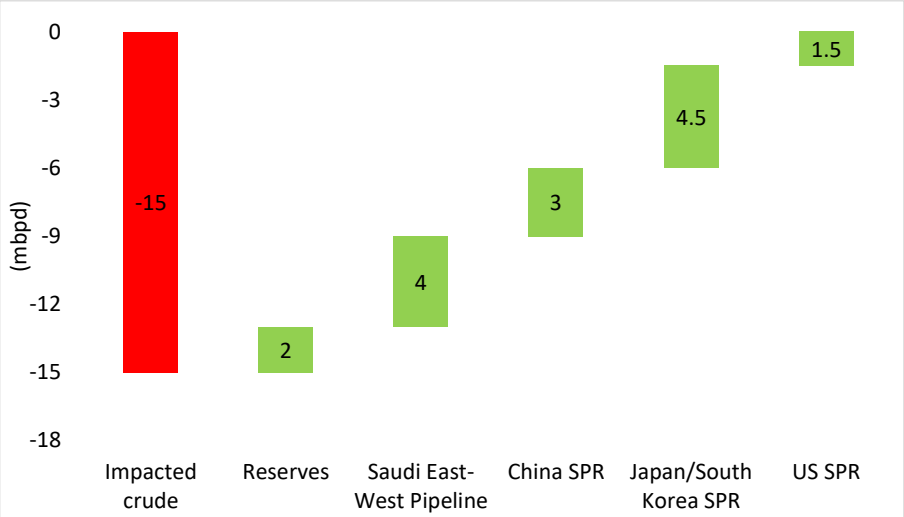
Strategic stocks to help bridge the supply gap

Of the ~15mbpd of crude moving through Hormuz impacted, roughly 3.5–4mbpd can still be routed through Saudi Arabia's east-west pipeline. Since the market also entered the crisis with something like a 2mbpd surplus, the effective net crude problem may be closer to 9mbpd rather than the full 15mbpd, and much of that shortage would be for Asia. China could cover a large part of its exposure through its own strategic petroleum reserves, which could reduce the remaining problem materially. Japan and South Korea also have substantial strategic stocks and meaningful daily drawdown capability. FGE's research shows that both Japan and South Korea have the potential to draw down from strategic stocks at a rate of over 4mbpd (although this has never been tested), while the US might manage around 1–1.5mbpd.

Compulsory stocks are an important part of the picture, because if governments allow those to be used, the oil is already within the system and can flow directly into consumption without the logistical complication of moving barrels from separate strategic sites.

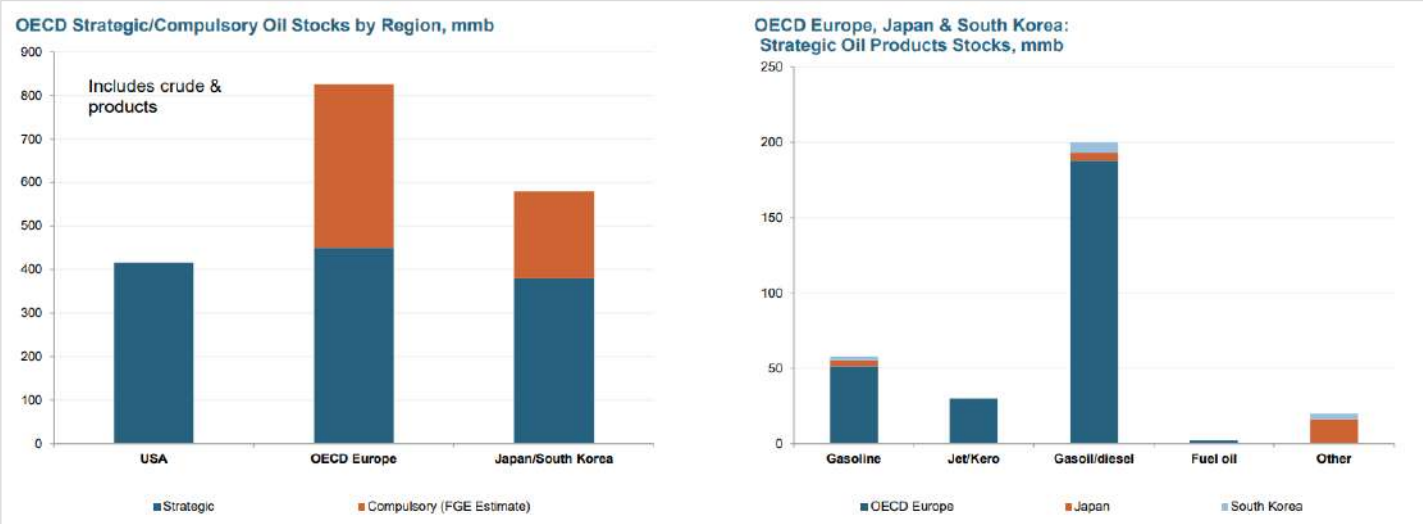
OECD countries collectively hold around 1.8bn barrels of strategic and compulsory stocks, which could cover roughly four to six months even before factoring in China. Including China, the system could potentially withstand the disruption for around a year.

Exhibit 3: Strategic reserves may help bridge the shortfall



Source: FGE NexantECA, Nuvama Research

Exhibit 4: Strategic reserves by region



Source: FGE NexantECA

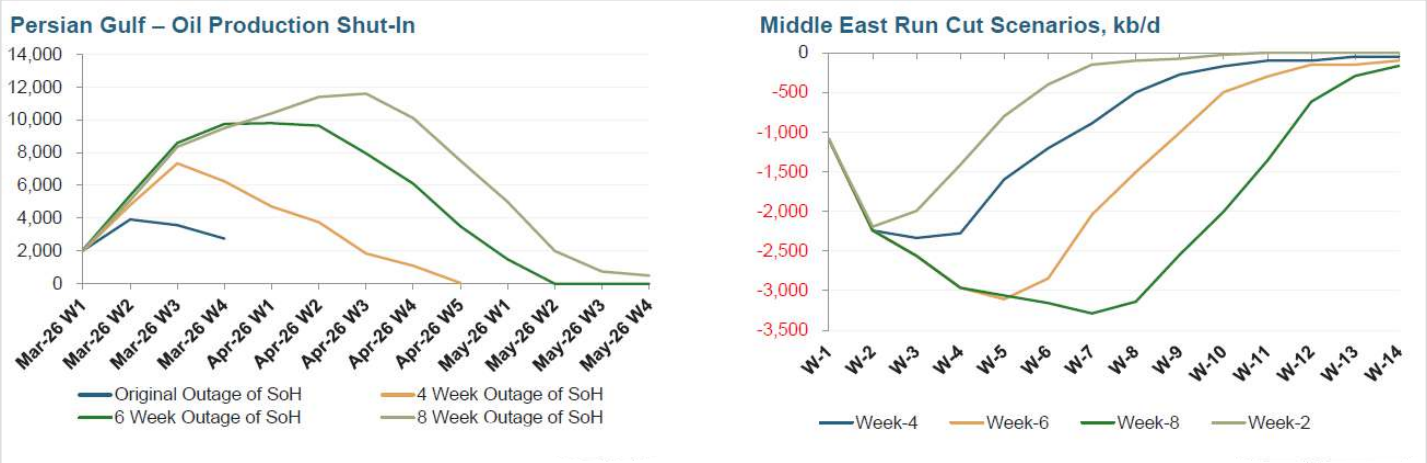
Policymakers are considering a strategic reserve release of ~300–400mn bbl to offset the shortfall, but this would only ease near-term pressure while creating future demand when inventories are replenished. Meanwhile, export constraints could force Middle East production shut-ins of ~6–7mbpd in March (>200mn bbl lost), keeping the crude market structurally tight even if emergency reserves are deployed.

Loss of crude flows may force refinery run cuts

FGE indicated that the loss of Middle Eastern crude flows might force refinery run cuts as planned crude arrivals fail to materialize and crude from alternative sources such as the US cannot arrive quickly enough. As a result, FGE estimates refinery shut-ins of ~2.5–3mbpd in March, leading to a ~10–15% reduction in Asian product flows and exacerbating the loss of product exports from the Persian Gulf. This could result in product output losses of ~600kbd of gasoline and ~900kbd of diesel in March, with disruptions persisting into April.

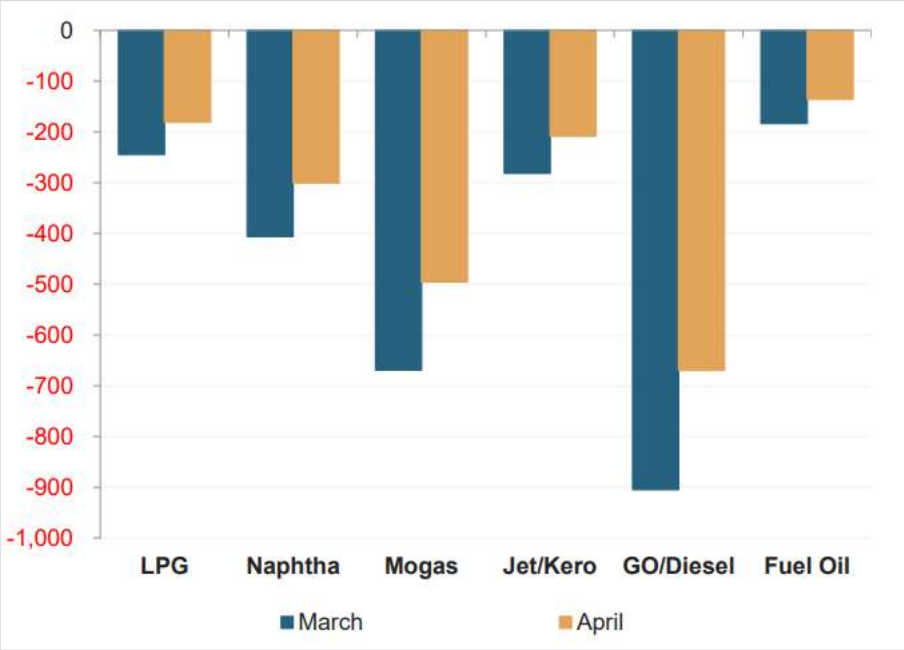
Refining margins have surged as the Middle East and Asian refinery runs drop ~2–2.5mbpd, with Europe adding only ~300–500kbpd and US refineries already highly utilized, leaving global capacity unable to offset disruptions. The Asian margin is USD5–7/bbl above end-February despite higher crude and freight costs.

Exhibit 5: Middle East shut-ins



Source: FGE NexantECA

Exhibit 6: Asian precautionary run cuts (kbpd)

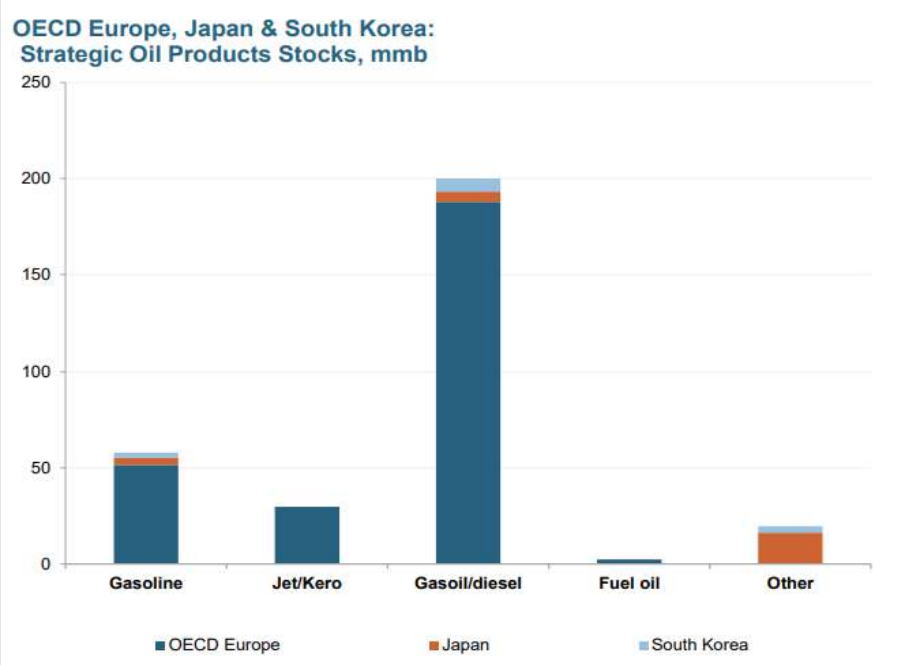


Source: FGE NexantECA

Petroleum products: Strategic reserves to help rebalance flows

OECD's strategic petroleum reserves (~1bn barrels) and Europe's diesel stocks (~180mn bbl) can help rebalance flows, enabling India to redirect diesel to Asia, while ATF remains vulnerable due to limited storage and straight-run production, keeping premiums over diesel elevated through summer and potentially winter. Strategic stock releases may stabilize the market, but strong margins are expected to persist into summer.

Exhibit 7: Europe holds 180mn bbl of diesel stock

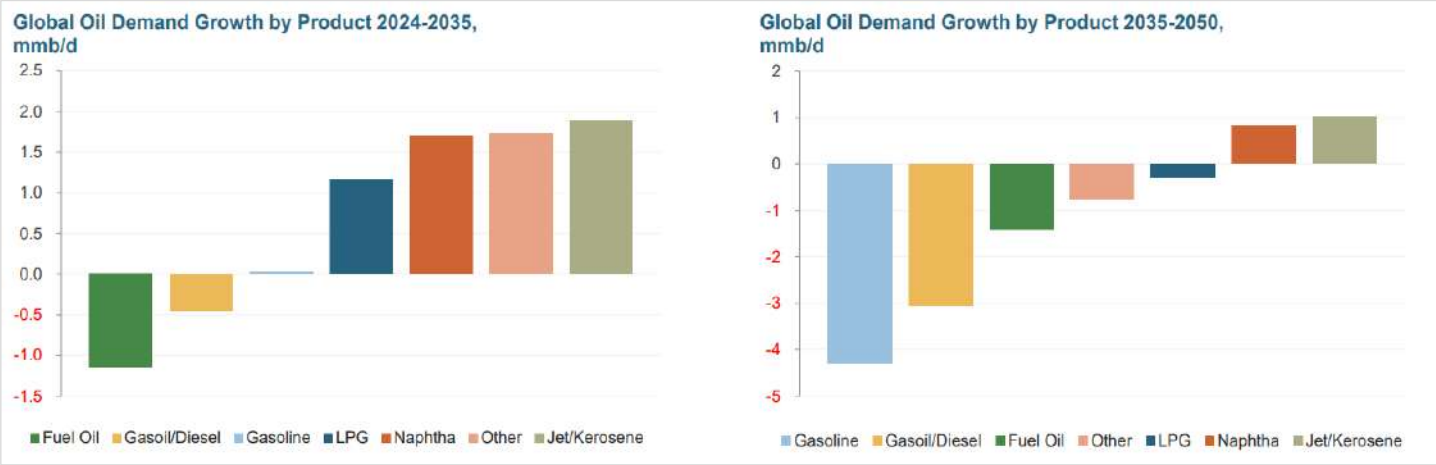


Source: FGE NexantECA

Oil products to see demand rise by 2035

Until 2035, all main products will see demand increase, apart from gasoil/diesel and fuel oil. Demand growth will be led by naphtha, followed by jet/kerosene. LPG will contribute strongly, driven by significant petrochemical capacity additions. After 2035, the reduction in road fuel demand will eat into gasoline and diesel use. Petrochemicals and jet fuel demand growth will continue, but at a slower pace.

Exhibit 8: Global oil demand outlook; changes by main product



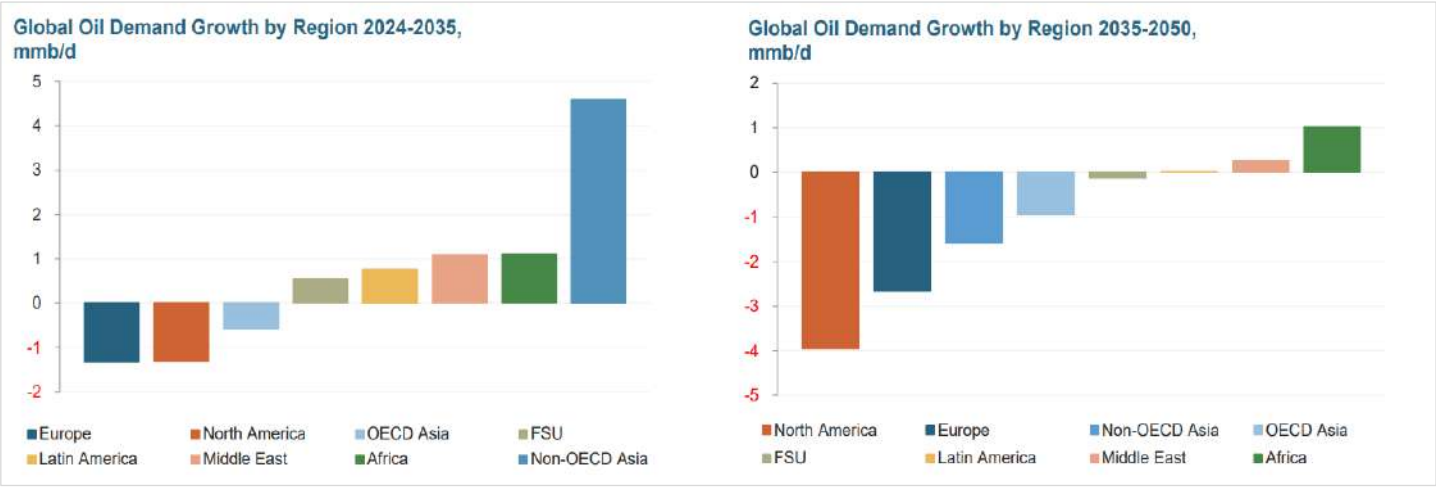
Source: FGE NexantECA

After 2035, naphtha and jet/kerosene will be the only products that will see demand growth, while gasoline and gasoil/diesel demand will decrease by 7.3mbpd combined.

Global oil demand outlook; changes by region

During 2024–2035, we project that oil demand in Europe, North America and OECD Asia will decrease by 3mbpd, while demand in non-OECD Asia will rise by about 4.6mbpd. Over 2035–2050, OECD oil demand is projected to decrease by 7.5mbpd. As Chinese demand is set to peak between 2030 and 2035, we expect non-OECD Asian oil demand to start decreasing after 2035, falling by 1.6mbpd between 2035 and 2050.

Exhibit 9: Africa and the Middle East are the only regions with a higher projected oil demand in 2050 compared with 2035



Source: FGE NexantECA

Refining: Long-term margins robust

Long-term outlook: Product demand to remain concentrated in Asia

Over the longer term, FGE expects global product demand growth through 2030 to remain concentrated in Asia, particularly in jet fuel and light-end products such as LPG and naphtha.

Exhibit 10: Asia's total oil demand growth (2024–2026), (mbpd)

Source: FGE NexantECA

EV adoption eroding gasoline demand

Electrification-driven by China's large-scale EV production and growing adoption across Asian markets such as Thailand, Vietnam and Indonesia is gradually eroding gasoline demand, though the structural impact on oil demand is likely to emerge only after ~2035.

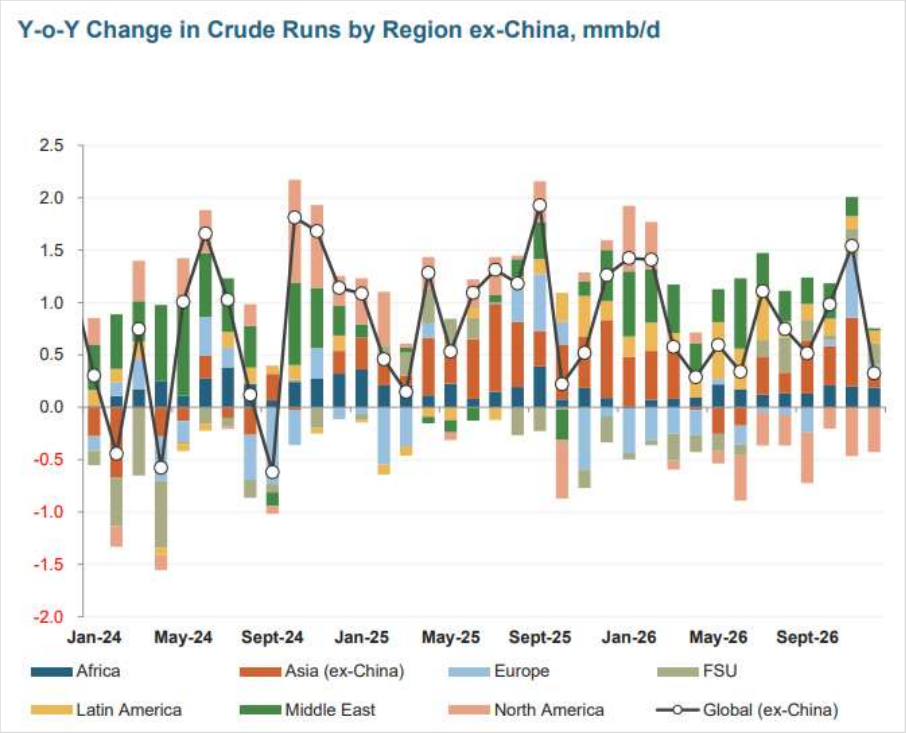
Exhibit 11: The top two EV markets to watch in Asia

Source: FGE NexantECA

Medium-term outlook: Capacity addition remains supportive

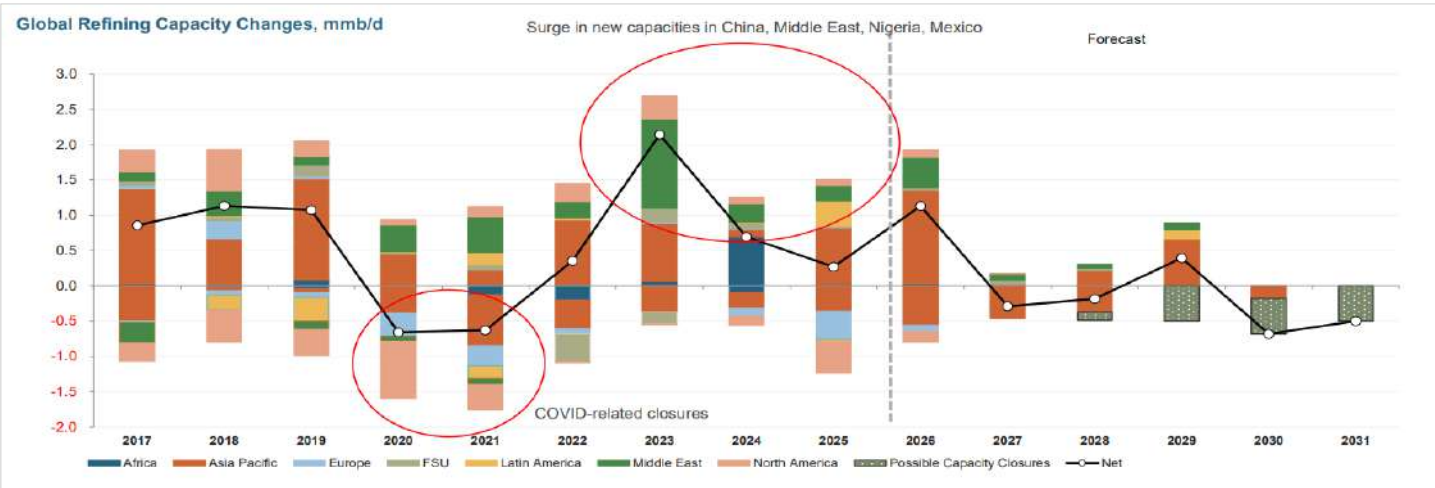
Over the medium term, jet fuel demand remains one of the strongest growth segments. On the supply side, refinery capacity addition remains supportive in the near term, with over 1mbpd of capacity expected in 2026 driven by projects such as India's Barmer refinery and China's Liaoning refinery (excluding China from some datasets), while capacity growth slows significantly thereafter with limited addition in 2027–28 before picking up again around 2029 (e.g., Fujian refinery). Expansion projects, particularly in India, are expected to support capacity growth as the country continues to expand refining to meet rising domestic demand and improve product self-sufficiency.

Exhibit 12: Capacity additions to pick from 2025 lows



Source: FGE NexantECA

Exhibit 13: Refinery capacity additions offset by closures ahead



Source: FGE NexantECA

Petrochemical chain under pressure

Petrochemical markets are also under pressure as naphtha-to-ethylene margins have turned negative, well below the ~USD250/t premium needed to justify integrated operations, implying either lower naphtha prices relative to crude or higher ethylene prices as steam cracker run cuts emerge. The loss of ~1.2mbpd of LPG and ~1mbpd of naphtha feedstock further tightens the petrochemical chain and could reduce supply of gasoline blending components such as pyrolysis gasoline (pygas) and aromatics (e.g., toluene and paraxylene).

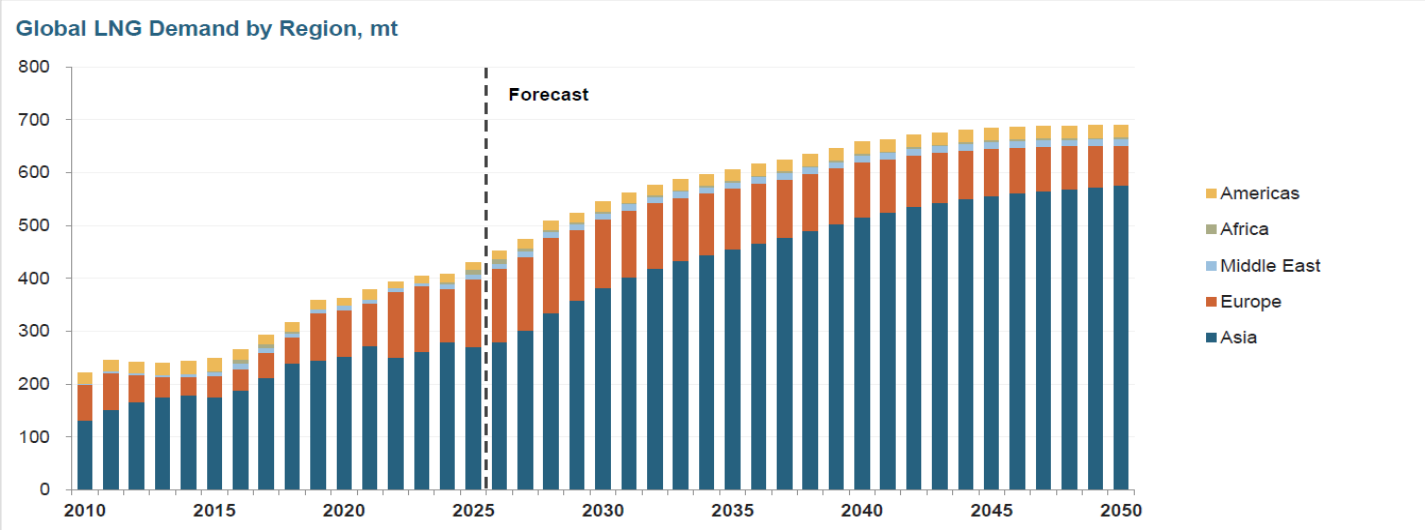
Freight costs rise on the back of Hormuz chaos

Freight markets have also tightened as trade flows adjust, with gasoline potentially moving from Europe to Asia due to open arbitrage opportunities (e.g., Singapore gasoline trading at a premium to Europe). At the same time, tanker availability has tightened, as vessels remain trapped in the Gulf, pushing both clean and dirty freight rates higher.

LNG: short-term shock, long-term growth

FGE highlighted the war and potential blockade of the Strait of Hormuz, through which ~80mtpa of LNG from Qatar and Abu Dhabi transit, could significantly disrupt markets, especially in Asia (India, China, Taiwan, Pakistan, Bangladesh), while Europe is less affected due to US Gulf Coast LNG. A daily blockade may cause loss of three LNG cargoes (210,000–230,000 tons). Markets may manage a four-week disruption, with rising spot prices and limited inventories in Japan, Korea and China providing some cushion.

Exhibit 14: Global LNG demand to remain strong over the long-term



Source: FGE NexantECA

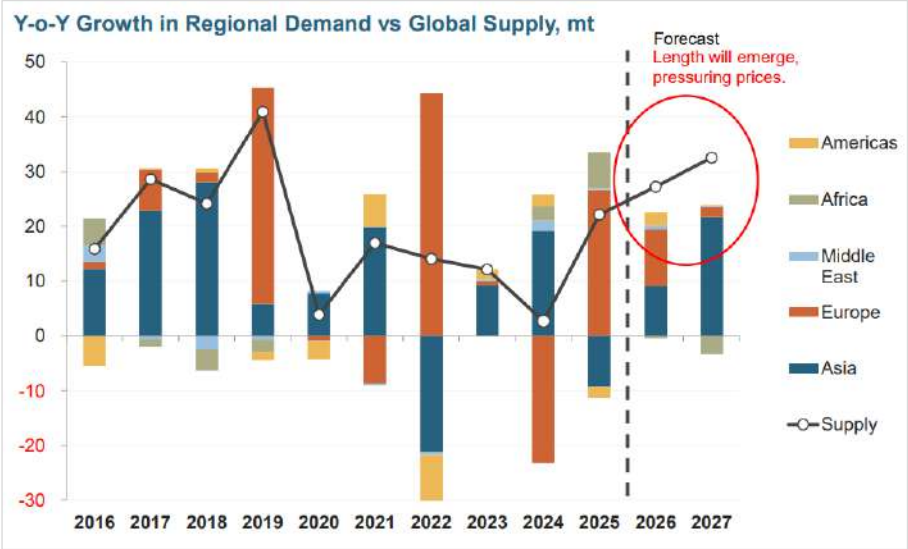
Asian demand to define 2026 market sentiment

Southeast Asian countries are securing cargoes and adjusting domestic energy use, while price-sensitive markets like India, Pakistan and Bangladesh may reduce imports, balancing supply at higher prices. Additional shocks, such as the shutdown of gas fields in Egypt, could increase spot LNG demand. Qatar's Ras Laffan facilities remain operational despite regional attacks, supporting recovery if no physical damage occurs.

Exhibit 15: Asia's engines of long-term LNG growth

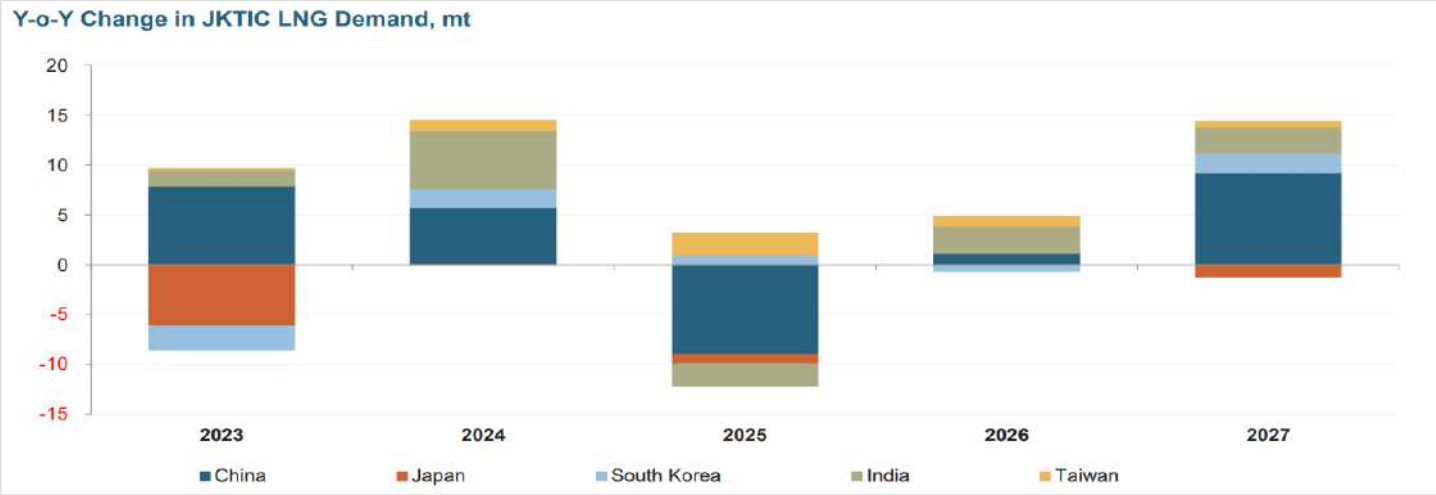
Source: FGE NexantECA

Exhibit 16: Asian LNG demand will be key in absorbing LNG supply growth



Source: FGE NexantECA

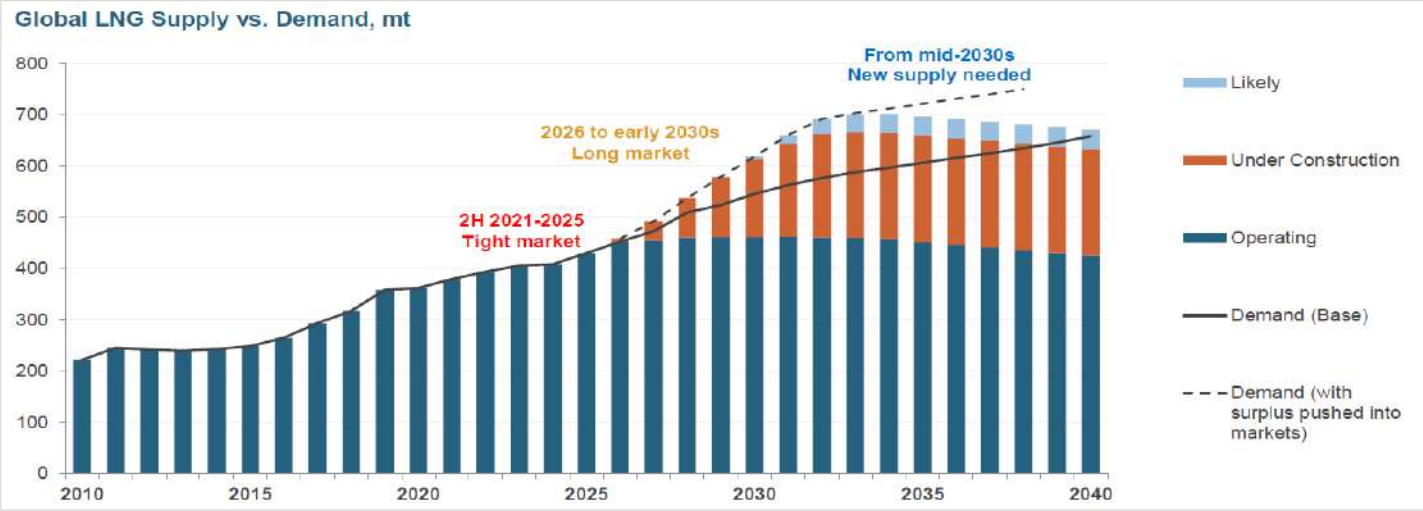
Exhibit 17: Asian LNG demand rebound expected in 2026 as more supply comes into the market



Source: FGE NexantECA

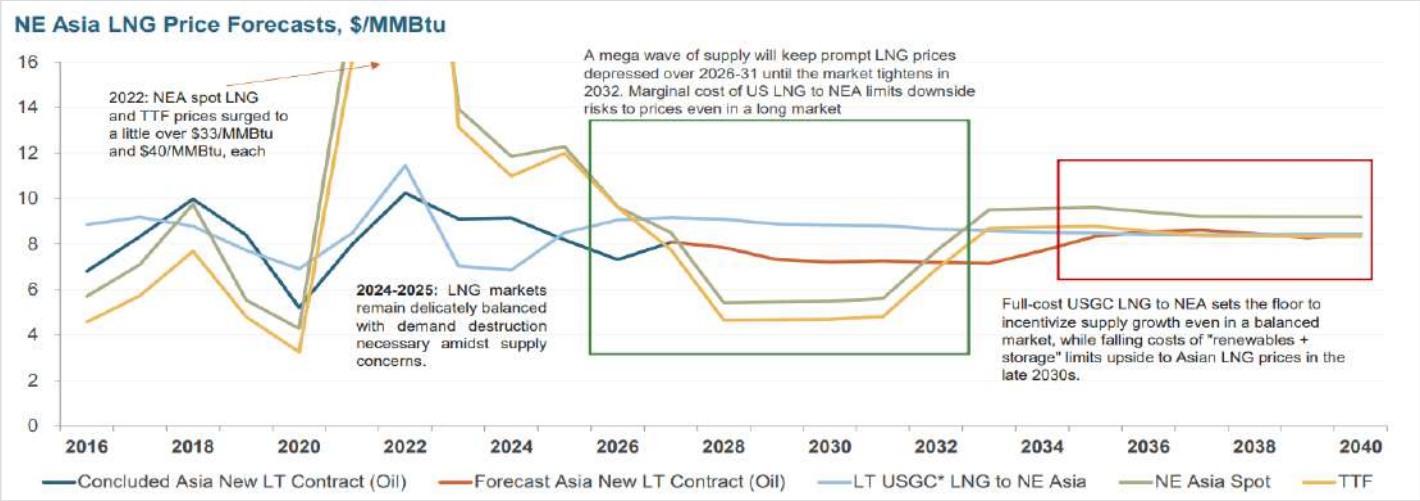
In the long term (2026–2030s), under-construction liquefaction capacity (230+mtpa for a 480mtpa market) ensures balanced supply, with incremental additions in 2027–2028 (~45mtpa) pressuring spot prices but maintaining a healthy market. Asia, particularly India, drives demand growth, supported by declining domestic production in Southeast Asia and rising imports, while planned projects are ready to meet further growth if geopolitical tensions ease, allowing a relaxed market through the early 2030s.

Exhibit 18: LNG market sees boom & bust cycles, with the tight market rebalancing in 2026



Source: FGE NexantECA

Exhibit 19: Long-term LNG prices converge around USD8–10/mmbtu



Source: FGE NexantECA

Implications on India

India is one of the most exposed countries to the Strait of Hormuz disruption across crude, LNG, and especially LPG. The common thread is that the risk is not limited to mark-to-market price movement; it is translating into real domestic shortages, operational disruption and economic stress. The overall picture is therefore mixed. Crude appears bridgeable over the short term through reserves and alternate sourcing. LPG is the hardest problem because there are no reserves. LNG remains manageable only up to a point, with affordability and demand destruction likely to become increasingly important if disruption persists.

Crude & Refining: India among the most exposed

- India is among the most exposed Asian buyers to the Hormuz disruption, alongside China, Japan and South Korea. A meaningful part of India's crude slate is linked to flows through the Strait of Hormuz, leaving it vulnerable to any prolonged blockage. Near-term buffers do exist in the form of strategic reserves, which can soften the initial crude shock.

- The crude shortfall appears bridgeable, but only through a mix of measures rather than a single replacement source. The balancing channels discussed include higher intake of fuel oil already available in the region as a partial substitute, increased Russian purchases under US dispensation, likely inflow of Venezuelan crude already at sea, domestic strategic reserve drawdown, with some lag, and higher Atlantic Basin flows moving west-to-east. It also depends on Reliance's SEZ refinery restarting or expanding Russian crude processing.

- India may also contribute to regional product balancing. With Europe holding substantial diesel inventories, India could divert some diesel volumes that would normally head to Europe towards deficit Asian markets such as Australia. As a result, India remains both a stressed buyer and a relevant balancing supplier in select product markets.

- Beyond the immediate crisis, India remains central to the medium-term refining growth story. The Barmer refinery is a key 2026 addition and India also features prominently in the 2026–27 expansion pipeline. The structural theme remains intact: India is still a growing oil-products market and is simultaneously seeking greater self-sufficiency, making it one of the main global drivers of refining capacity growth.

India's LPG vulnerability

- LPG is India's most acute near-term vulnerability. Unlike crude, diesel, and jet, LPG is not meaningfully covered by global strategic reserve systems. OECD's strategic and compulsory stocks contain only a very small LPG cushion, with Japan accounting for most of it. This leaves limited practical reserve-based relief for India's cooking gas shortage.

- The risk is physical, not just price-led, the key issues being insufficient cooking gas availability and warning of restaurant shutdowns, as the disruption feeds directly into the real economy.

- Balancing on LPG may partly come through demand destruction, with some segments forced to curtail consumption until supply conditions improve. A secondary relief mechanism could come from reduced steam-cracking activity, including LPG-fed crackers, which may free up some LPG and naphtha molecules

for redirection into markets such as India. However, this is only a partial offset rather than a full solution.

- Under the current working assumption of around four weeks of Hormuz-related disruption, followed by additional time for refining systems to normalize and cargoes to arrive, India's LPG stress could take roughly six to eight weeks to ease meaningfully. Domestic cooking-gas tightness may therefore outlast the initial geopolitical shock.

India's high reliance on Middle Eastern LNG supply

- The LNG issue is also feeding into the physical economy. Factory disruptions due to inadequate natural gas availability are reinforcing that the stress is operational rather than merely financial.
- India sits firmly in the group of price-sensitive LNG buyers. If spot LNG prices rise sharply, import reduction becomes a likely balancing response. In practice, a part of the market adjustment may therefore come through lower India demand, industrial curtailment and weaker downstream consumption rather than complete replacement of all lost volumes.
- Despite the short-term strain, India remains a major long-term LNG growth market. The demand case is still supported by coal-to-gas switching, rising energy demand and declining domestic gas supply, all of which point to higher import dependence over time.
- Affordability remains a core risk variable. Even if LNG is physically available, India may not absorb very high spot prices smoothly, making price tolerance just as important as supply access in assessing the country's LNG vulnerability.

Q&A session

Could the prolonged period of higher oil and gas prices create a new energy moment similar to what happened in the early 2000s, when high conventional energy prices helped create room for alternatives such as EVs, renewables, shale, green hydrogen and green ammonia to become more viable and attract investment?

Before the crisis the oil market had been looking oversupplied by around 2–4mbpd, which would likely have led to a lower oil price environment of roughly USD50–70/bbl over the next one to two years and would have hurt green investments. The crisis changes that because the short-term oil loss has to be paid back later, raising apparent demand further out. That could leave prices elevated for the next 12–18 months, in a USD70–80/bbl range after the immediate disruption fades. However, this higher price deck is more likely to stimulate tighter and quicker conventional supply responses, especially US tight oil, than to immediately unlock green investment at scale. Roughly 500kbpd of US tight production from high-cost producers had been lost over the prior 12 months and could return, taking US liquids output back to around 14–14.2mbpd.

In the longer run, the crisis does reinforce the security-of-supply case for moving away from dependence on vulnerable hydrocarbon routes. However, the actual transport transition still unfolds slowly, with structural impact showing up more around 2035. On green hydrogen and similar technologies, progress is still highly policy-driven, and recent policy trends have been more about delay than acceleration as the cost-of-living crisis makes politicians more likely to ease the burden on consumers, for example by reducing taxes on transport fuels rather than force faster decarbonisation in the short term.

If the Strait of Hormuz stayed closed and supply losses reached around 5mbpd and kept rising, how much of the shortfall could be offset by strategic oil reserve releases in Europe and the US, along with a possible lifting of sanctions on Russian oil?

The answer depends on whether one is measuring the total volume of reserves available or the physical release rate that can be sustained day-by-day. Starting with crude, ~15mbpd of crude normally moving through Hormuz, roughly 3.5–4mbpd can still be routed through Saudi Arabia's east-west pipeline. Since the market also entered the crisis with something like a 2mbpd surplus, the effective net crude problem may be closer to 9mbpd rather than the full 15mbpd. Much of that shortage would need to be solved in Asia.

China could cover a large part of its exposure through its own strategic petroleum reserves, which could reduce the remaining problem materially. Japan and South Korea also have substantial strategic stocks and meaningful daily drawdown capability. FGE's research shows that both Japan and South Korea have the potential to draw down from strategic stocks at a rate of over 4mbpd (although this has never been tested), while the US might manage around 1–1.5mbpd. Compulsory stocks are an important part of the picture, because if governments allow those to be used, the oil is already sitting inside the system and can flow directly into consumption without the logistical complication of moving barrels from separate strategic sites.

OECD countries collectively hold around 1.8bn barrels of strategic and compulsory stocks, which could cover roughly four to six months even before factoring in China.

Including China, the system could potentially withstand the disruption for around a year.

If spot LNG prices rise sharply enough, a similar sanction-relief logic could emerge in LNG. From Russia's Arctic 2 LNG project, two trains are ready to operate at reasonable capacity but are constrained by sanctions. At present, sanctioned cargoes have effectively only gone to China via Beihai terminal, but if there were temporary sanction exemptions, some cargoes from Arctic 2 and smaller Russian LNG projects could reach the market and slightly offset lost Qatari and Abu Dhabi LNG.

Where will India source crude from and why can't reserves easily solve the LPG issue too? Are those alternative sources really enough to plug a roughly 3mbpd gap?

India had already been taking up more fuel oil available in the region, effectively substituting part of its crude requirement with fuel oil. India had special dispensation from the US to buy more Russian oil. An open question remains as to whether Reliance's export-oriented SEZ refinery resumes processing more Russian crude. In addition, Venezuelan crude is still a few million barrels on the water that had not been sold and that these could help India in the short term.

The scale of the problem is enormous. The key is that the market is not trying to solve a permanent 3mbpd gap immediately; it is trying to bridge March and perhaps April. Beyond that, Atlantic Basin crude flows should start increasing into Asia. The high freight rates had initially hampered that response, but freight began easing.

~25 empty VLCCs that were waiting to enter the Gulf had instead passed Bab el-Mandeb and were now dispersing westward, which should help restart a stronger west-to-east crude flow. India's willingness to bid fuel oil to very high premiums even for the second month forward shows how acute the immediate shortage is. March and April are the key bridge months, after which other flows may increasingly come into play.

Short-term disruptions should be solved as absolute volume gaps rather than as ongoing daily deficits. For example, if India faces a 3mbpd issue for 30 days, the problem is 90 million barrels. If India can cover, say 30 million barrels from its own strategic reserves, and then the remaining problem is 60 million barrels, which can then be chipped away using imports from alternate sources.

When does India actually get back to normal, especially for LPG shortages affecting the real economy?

The LPG issue is especially problematic because strategic reserves are not really designed for it. Among OECD countries, only Japan has any meaningful LPG in strategic storage. There are roughly 18 million barrels of LPG across OECD's strategic and compulsory stocks in total, of which around 15 million barrels are in Japan. The global strategic stock system was designed around diesel, jet, and gasoline, not LPG and naphtha, partly because the petrochemical sector was never really built into that reserve framework. As a result, India cannot count on a large strategic LPG safety net.

Regarding LPG, the solution depends on some restoration of Hormuz flows plus time for the refining system to catch up. Under the base assumption of four weeks of disruption, followed by another two to three weeks for refining and cargo delivery normalization. The effective time needed for the resumption of more normal conditions could be around six to eight weeks.

One balancing factor is that steam-cracking operations, including LPG-fed and flexi-feed crackers, are being cut significantly. That should free up at least some LPG volumes that can be redirected toward India. However, some demand destruction may be unavoidable. The industry has yet not seen the end of steam-cracker reductions.

About FGE

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About Mr James Davis (Director, FGE)

Mr James Davis leads FGE's Short-Term service and crude oil production forecasting. He also covers crude oil price differentials analysis and regularly provides input into FGE's short-term market research reports. Since joining FGE in 2013, Mr Davis has upgraded the group's supply-side modelling capabilities providing detailed field-by-field production profiles, which forms an integral part of the group's short and long term research. He is an M.Eng in Civil Engineering from the University of Bristol. Prior to this, he was employed as a corporate finance analyst in the upstream oil & gas sector, working with small to mid–cap E&P companies.

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Eugene is Head of Refined Products at FGE. He has dedicated the last 15 years to cross-barrel oil analysis. At different points of his career he headed both the Downstream and Upstream research teams at JBC Energy, one of the leading oil and gas market research centres. Eugene has used his vast knowledge of energy markets to advise national and international oil and gas companies as well as government-related agencies around the world for many years. Being an oil pricing expert, he has supported some of the world's largest oil producers in setting their monthly selling prices. Eugene also played a decisive role in shaping an industry-leading view on the global impacts of IMO 2020, presenting these views at conferences across the globe. Prior to his work at JBC Energy, Eugene spent several years working in an intelligence cell at NATO regional headquarters in Netherlands.

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